3.8 Bivariate Frequency Distributions

It is known that the frequency distribution of a single variable is called univariate distribution. When a data set consists of a large mass of observations, they may be summarized by using a two-way table. A two-way table is associated with two variables, say X and Y. For each variable, a number of classes can be defined keeping in view the same considerations as in the univariate case. When there are m classes for X and n classes for Y, there will be m × n cells in the two-way table. The classes of one variable may be arranged horizontally, and the classes of another variable may be arranged vertically in the two way table. By going through the pairs of values of X and Y, we can find the frequency for each cell. The whole set of cell frequencies will then define a bivariate frequency distribution. In other words, a bivariate frequency distribution is the frequency distribution of two variables.

Table 3.17 shows the frequency distribution of two variables, namely, age and marks obtained by 50 students in an intelligent test. Classes defined for marks are arranged horizontally (rows) and the classes defined for age are arranged vertically (columns). Each cell shows the frequency of the corresponding row and column values. For instance, there are 5 students whose age fall in the class 20 – 22 years and their marks lie in the group 30 – 40.

Table 3.17

Bivariate Frequency Distribution of Age and Marks

Marks	Age in Years				Total
	16 – 18	18 - 20	20 - 22	22 – 24	
10 – 20	2	1	1	-	4
20 – 30	3	2	3	1	9
30 – 40	3	3	5	6	17
40 – 50	2	2	3	4	11
50 – 60	-	1	2	2	5
60 – 70	-	1	2	1	4
Total	**10**	**10**	**16**	**14**	**50**

3.9 Stem and Leaf Plot (Stem and Leaf Diagram)

The *stem* and *leaf* plot is another method of organizing data and is a combination of sorting and graphing. It is an alternative to a tally chart or a grouped frequency distribution. It retains the original data without loss of information. This is also a type of bar chart, in which the numbers themselves would form the bars.

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